

Department of Computer Science and Engineering(CSE)

INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG(IIUC)

Course code:CSE-3524

Course Title:Microprocessor & Microcontroller Sessional

EXPERIMENT 2

Experiment Name: Familiarization with PUSH POP Operation, loading the program code and verification of program with it.

OBJECTIVES:

1. To write a program for exchanging numbers using PUSH/POP.
2. To generate the list file.
3. To observe the equipments of kit box.
4. To check the register values in kit box.
5. To analyze the result.

NECESSARY EQUIPMENT:

1. Emu8086 software to run the program.
2. Notepad to write the program.
3. 8086 microprocessor kit.

PROBLEM STATEMENT:

Write an assembly language program to load 16 bit Hex number in two general purpose registers. Interchange their contents using PUSH-POP instruction. Take the initial address, ,SP= 2009H.

Exp2: Interchanging the contents of two register using PUSH-POP;

```
.MODEL SMALL  
.CODE
```

```
MOV SP, 2009H  
MOV AX, 5566H  
MOV BX, 7788H  
PUSH AX  
PUSH BX  
POP AX  
POP BX  
END
```

DATA TABLE:-

Instruction	AX	BX	CX	DX	SP	Set Flag
MOV SP, 2009H						
MOV AX, 5566H						
MOV BX, 7788H						
PUSH AX						
PUSH BX						
POP AX						
POP BX						

REPORT WRITING:-

1. Include experiment name, objectives, necessary software/ equipment and problem statement.
2. Insert the printed copy of “list file” of the program.
3. Draw the relevant memory maps.
4. Find the result of all related registers from kit box.
5. Write discussion if you have any error in the program or troubleshooting of the program.
6. Write the answer of the related questions given below.

RELATED QUESTIONS:-

1. Can you interchange the contents of two registers with another way?
2. If the answer of question1 is “yes” then write at least two new ways (Assembly Language Program) for interchanging two register contents.

List file:

EMU8086 GENERATED LISTING. MACHINE CODE <- SOURCE.

noname.bin -- emu8086 assembler version: 4.08

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=====		
=====		
[LINE]	LOC: MACHINE CODE	SOURCE
=====		
=====		

[1]	:	
[2]	0000: BC 09 20	MOV SP, 2009H;
[3]	0003: B8 66 55	MOV AX, 5566H;
[4]	0006: BB 88 77	MOV BX, 7788H;
[5]	0009: 50	PUSH AX ;
[6]	000A: 53	PUSH BX;
[7]	000B: 58	POP AX ;
[8]	000C: 5B	POP BX ;
[9]	:	END